Search the web for established benchmarks and metrics used in scientific papers to evaluate multi-agent LLM systems, specifically:

1. Metrics for measuring "context collapse" or "coherence degradation over time" in multi-agent LLM debates/conversations
2. Metrics for measuring "resilience" of multi-agent systems - ability to sustain quality actions over time
3. Metrics for measuring "autonomy" in agent-based systems
4. Established benchmarks like tau-bench, AgentBench, LMRL-Gym, SWE-bench, or others used in recent (2024-2025) multi-agent evaluation papers

For each metric/benchmark found, report:

- Name and citation
- What it measures
- Whether it captures temporal degradation (quality over time, not just average)

Return a concise list of the most relevant options.

Here is a concise list of metrics and benchmarks that map well to your four targets, emphasizing those that explicitly consider time / interaction length.

1. “Context collapse” / coherence degradation over time

a. CORE (Conversational Robustness Evaluation)

- **Citation:** “CORE: Measuring Multi-Agent LLM Interaction Quality under Game-Theoretic Dynamics” (EACL 2026).[1]
- **What it measures:** A single CORE score in $[0, 1]$ that combines three dialog-level phenomena in multi-agent interactions:
  - Mode collapse (agents converging to a narrow set of responses),
  - Lexical repetition,
  - Semantic stagnation (no new information / ideas emerging).[1]
- **Temporal degradation?** Yes. The metric is defined over *multi-turn* multi-agent dialogs, and the components explicitly track diversity and semantic change across turns, so it directly captures degradation of conversational richness as the interaction progresses, not just a per-turn average.[1]

b. Multi-turn dialog robustness metrics (surveyed)

- **Citation**: "Evaluating LLM-based Agents for Multi-Turn Conversations: A Survey" (2025). [2]
- **What they measure**: The survey catalogs metrics used for multi-turn agents, including: turn-level coherence, topic consistency, persona consistency, error propagation, and context retention across turns.[2]
- **Temporal degradation?** Yes in spirit, but mostly via *derived curves* rather than explicit "collapse" scores: e.g., measuring quality at each turn and analyzing decline, or measuring the fraction of turns that remain on-topic as depth increases. The survey notes that current metrics "struggle to capture" interaction-level degradation, which is partly why frameworks like CORE are being proposed.[2] [1]

c. MultiAgentBench / MARBLE coordination degradation with iterations

- **Citation**: MultiAgentBench + MARBLE framework, "MultiAgentBench: Evaluating the Collaboration and Competition of LLM agents" (ACL 2025).[3]
- **What it measures**: Their core metrics (Task Score TS, Communication Score, Planning Score, Coordination Score, KPI) are defined over multi-turn interactions in various scenarios (research, Minecraft, database debugging, coding, bargaining, Werewolf).[3]
- **Temporal degradation?** Indirectly, yes. They run explicit ablations over the **maximum number of interaction iterations** and plot TS and coordination metrics vs iterations; they show that beyond ~7 iterations, coordination degrades even if task scores partially recover. This gives you a pattern you can reuse: treat TS/CS as a function of turn/iteration and measure degradation (e.g., slope or late-stage vs early-stage gap).[3]

d. Self-evolving / robustness benchmarks (single/multi-agent)

- **Citation**: "Self-Evolving Benchmark" (2024) as summarized in "Evaluating LLM-based Agents: Metrics, Benchmarks, and Best Practices." [4]
- **What it measures**: Robustness of agents when inputs are perturbed (paraphrases, noisy contexts, extended contexts). Metrics include original task accuracy vs performance on perturbed/evolved instances, and fine-grained measures such as change in chain-of-thought quality.[4]
- **Temporal degradation?** Mostly **input-variation** rather than step-wise temporal; but conceptually close to measuring **context sensitivity** and context overload. You could adapt the "performance drop between original vs perturbed" idea across turns.

2. "Resilience" of multi-agent systems over time

a. τ-bench (tau-bench): pass^k reliability metric

- **Citation**: Yao et al., "τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains" (2024).[5] [4]
- **What it measures**: A tool-using agent interacting with a (simulated) user under domain-specific tools and policies. Core metrics:

    - Task success (goal state of a database after the conversation).
    - **pass^k**: probability of success over multiple independent runs (e.g., pass^8) to measure reliability/consistency of behavior.[4]
- **Temporal degradation?** Partially. pass^k aggregates across runs, not within a single long run, so it captures **resilience under repetition** (can the agent consistently succeed when replayed), but not degradation *within* one long conversation. Still a useful precedent for "resilience = sustained success across trials."[4]

b. Cooperative resilience in AI multi-agent systems (continuous-time resilience metric)

- **Citation**: "Cooperative Resilience in Artificial Intelligence Multiagent Systems" (2024).[6]
- **What it measures**: A general **resilience functional** for multi-agent systems exposed to successive disruptive events (environment changes, adversarial agents). They:
    - Define performance and reference trajectories over time,
    - Segment into time windows around each disruption,
    - Compute windowed resilience by comparing performance vs reference,
    - Aggregate into a scenario-level resilience score J_k that penalizes decreasing resilience and rewards improvement across disruptions.[6]
- **Temporal degradation?** Yes, explicitly. Resilience is a **time-windowed integral / comparison** and final scores are sensitive to *sequences* of events and learning or degradation over consecutive disruptions. This is the closest thing to a general-purpose "resilience over time" construct you can reuse.[6]

c. Logic-Driven Semantic Communication for Resilient Multi-Agent Systems

- **Citation**: "Logic-Driven Semantic Communication for Resilient Multi-Agent Systems" (2026 preprint).[7]
- **What it measures**: Formal definitions and metrics of multi-agent resilience split into:
    - Epistemic-based resilience (how beliefs / knowledge states adapt under stressors),
    - Action-based resilience (how agents' actions maintain performance).[7]
      These metrics are defined within a *temporal epistemic* framework, so they reason about states and actions over time.
- **Temporal degradation?** Yes in the formalism: resilience is explicitly tied to temporal properties (what agents know and do across time under stressors). It is more theoretical than τ-bench but could be instantiated for LLM agents.[7]

d. MARBLE / MultiAgentBench iteration and disruption experiments

- **Citation**: MultiAgentBench paper (coordination protocols, planning strategies, ablations).[3]
- **What it measures**:
    - TS (Task Score) and CS (Coordination Score) as you vary interaction iterations and agent count.

- They show that **too many iterations** can cause coordination degradation and that adding more agents reduces KPI and complicates collaboration.[3]

- **Temporal degradation?** Yes, empirically. They treat performance metrics as a function of iteration count and use that to argue about coordination resilience (e.g., "excessive iterations may lead to coordination degradation"). You can reuse this pattern for your own resilience metric: e.g., area under TS curve vs turns or minimum TS beyond a certain depth.[3]

3. "Autonomy" metrics for agent-based / LLM systems

a. "Quantify Autonomy for Agent Based Systems" (software metrics)

- **Citation**: "Quantify Autonomy for Agent Based Systems" (serials journals technical report).[8]
- **What it measures**: Autonomy through observable attributes such as:
  - Agent communication patterns,
  - Process time, time to grasp surroundings, switching time between environments,
  - Action types and frequency, number of unpredictable environment changes handled,
  - Interoperability, belief/reputation dynamics, pro-activity, negotiation, task sharing, and role behavior.[8]
- **Temporal degradation?** Yes, potentially. Many of these metrics are **time-sensitive** (e.g., process time, switching time, handling rate of changes), so tracking them over episodes directly reflects degradation of autonomy under load or over long runs. The paper itself focuses on defining the attribute space rather than a single scalar score, but it gives a structured basis.[8]

b. Measuring AI Agent Autonomy via attribute-specific metrics (Impact / Oversight)

- **Citation**: Cihon et al., "Measuring AI agent autonomy: Towards a scalable approach with code inspection" (2025), summarized under "Attribute-Specific Autonomy" (Emergent Mind).[9]
- **What it measures**: A **measurement and scoring framework** that decomposes autonomy into attributes grouped under:
  - **Impact**: what actions the system may take, environmental scope, potential real-world effects.
  - **Oversight**: how interactions are orchestrated/bounded, when/how human input is sought, logging/observability.[9]
    The idea is to compute attribute-level scores (e.g., degree of human oversight required for specific functions).
- **Temporal degradation?** Not directly; these are *structural* autonomy metrics (based on capabilities and oversight mechanisms), not performance curves. You could, however, sample these scores across releases or configurations to see autonomy changes over time, but that is not how the original work uses them.[9]

c. Practical autonomy metrics in LLM agent surveys

- **Citation:** "Evaluating LLM-based Agents: Metrics, Benchmarks, and Best Practices" (2025 blog/survey).[4]
- **What it measures:** Proposes categories like robustness, reliability, and failure attribution. For agents, autonomy is operationalized via:
  - Degree of human intervention needed,
  - Fraction of tasks completed without manual correction,
  - Ability to self-correct and recover from errors across multi-step workflows.[4]
- **Temporal degradation?** Implicitly yes, if you track failure attribution and performance step-by-step within a run; the article advocates logging each agent step and attributing failures to specific agents / steps, which is effectively a temporal trace of autonomy failures.[4]

4. Multi-agent / agentic benchmarks (2024–2025) and what they capture

Below is a compact table focusing on your named benchmarks and close relatives.

Key multi-agent / agentic benchmarks (2024–2025+)

Benchmark / framework	What it measures	Temporal degradation?
τ-bench (tau-bench) (Yao et al., 2024) [5] [4]	Dynamic conversations between user (simulated) and tool-using agent with domain policies; task success via final DB state; pass^k for reliability over multiple runs.	Partially: pass^k captures sustained success across trials, but not explicit within-run quality decay.
MultiAgentBench + MARBLE (ACL 2025) [3]	Multi-agent collaboration and competition (research, Minecraft, database, coding, bargaining, Werewolf). Metrics: Task Score (TS), KPI (milestone contributions per agent), Communication Score, Planning Score, Coordination Score, Competition Score.	Yes, via ablations vs number of iterations and agents; they show coordination and TS as functions of interaction depth, revealing degradation with too many steps. [3]
AgentBench (2023, widely reused 2024–2025) [3]	Mostly single-agent but considered in multi-agent surveys; evaluates agents on diverse tool-use and interactive tasks (web, database, games, etc.) using success/accuracy metrics.	Not specifically; mostly episode-level success, though environment is interactive.
Self-Evolving Benchmark (2024) [4]	Dynamic generation of perturbed tasks (noise, paraphrases, out-of-domain twists) via a multi-agent "reframing" system; measures robustness as drop in performance on evolved vs original instances.	Indirectly: measures robustness under increasingly "hard" or noisy contexts, not turn-wise decay, but conceptually adjacent to context collapse.

Benchmark / framework	What it measures	Temporal degradation?
Game / social benchmarks (e.g., Werewolf, bargaining, GameNGen, CUISINEWORLD) (see MultiAgentBench related work) [3]	Evaluate emergent social behavior, negotiation, deception, and collaboration. Metrics: win-rates, negotiation efficiency, emergent behavior counts (e.g., strategic info sharing).	Often yes: many are multi-round games where outcomes depend on consistent strategy over turns. MultiAgentBench explicitly quantifies emergent behaviors over full transcripts. [3]
TAU-bench (BenchLM category) [10]	BenchLM lists TAU-bench under an "Agentic" category with an overall score aggregated over models.	This is a reporting/aggregation site; underlying temporal behavior is inherited from τ-bench itself. [10] [5]
LLM-RL / LMRL-Gym type environments (by analogy)	RL-style environments for LLM agents, typically measuring episodic return, success rates, sample efficiency; often used for agentic RL + LLM hybrids.	Yes, in RL terms: reward trajectories and performance vs episode length, but explicit "context collapse" metrics are not standard. (You would likely have to define your own temporal degradation analysis.)

Note: SWE-bench itself is single-agent code repair; multi-agent papers often use it as an underlying task but aggregate with standard accuracy metrics, not multi-agent specific ones.

How these map to your design needs

If you want to evaluate **context collapse / coherence over time** in multi-agent LLM debates, the closest ready-made pattern is:

- Use **CORE** or a similar diversity/semantic-change metric over **sliding windows of turns** to directly quantify mode collapse, repetition, and stagnation across the debate.[1]
- Combine with **turn-indexed TS / CS curves** as in MultiAgentBench to show how task/coordination quality changes after N turns.[3]

For **resilience**, you can:

- Use **τ-bench-style pass^k** over repeated runs to capture robustness to stochasticity.[4]
- Layer on **cooperative resilience curves** (performance vs reference trajectory over disruptions) from the resilience papers.[6] [7]

For **autonomy**, you could:

- Adopt attribute-level autonomy scores (Impact/Oversight) from Cihon et al.[9]
- Add **operational autonomy** metrics: fraction of tasks completed without human intervention, and rate of successful self-corrections per error, logged step-wise as suggested in the agent evaluation survey.[4]

Would you like a concrete metric set (formulas + logging schema) that you could drop into your own multi-agent debate framework to track collapse, resilience, and autonomy over long horizons?

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1. https://aclanthology.org/2026.eacl-long.57.pdf
2. https://arxiv.org/html/2503.22458v1
3. https://aclanthology.org/2025.acl-long.421.pdf
4. https://samiranama.com/posts/Evaluating-LLM-based-Agents-Metrics,-Benchmarks,-and-Best-Practices/
5. https://arxiv.org/abs/2406.12045
6. https://arxiv.org/html/2409.13187v2
7. https://arxiv.org/pdf/2601.06733.pdf
8. https://serialsjournals.com/abstract/25742_cha-3.pdf
9. https://www.emergentmind.com/topics/attribute-specific-autonomy
10. https://benchlm.ai/benchmarks/tauBench
11. https://www.confident-ai.com/blog/llm-evaluation-metrics-everything-you-need-for-llm-evaluation
12. https://www.lesswrong.com/posts/tGcLA596E8g3KnphE/survey-of-multi-agent-llm-evaluations
13. https://arxiv.org/html/2602.02597v1
14. https://wandb.ai/onlineinference/genai-research/reports/LLM-evaluation-Metrics-frameworks-and-best-practices--VmlldzoxMTMxNjQ4NA
15. https://www.evidentlyai.com/llm-guide/llm-evaluation-metrics